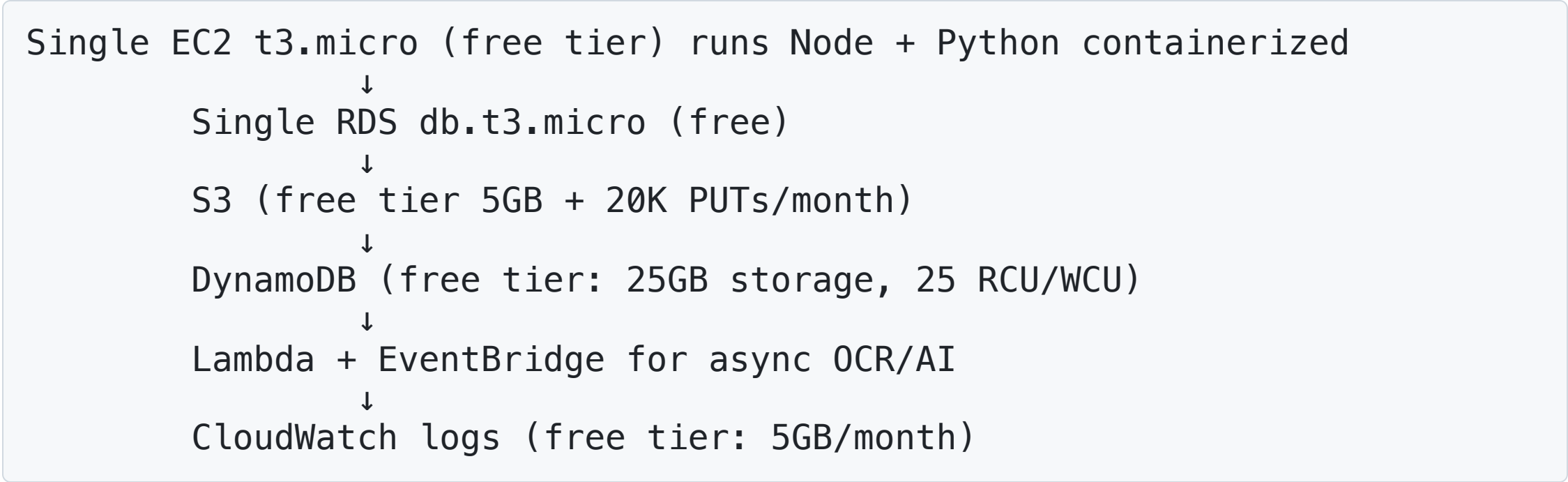


AWS Cost-Optimized Demo Setup

Goal: Ship a fully-featured POC for ~\$0–\$100/month instead of \$1,500+

Option 1: Serverless-First (Recommended for Demo)

Architecture



Monthly Cost Breakdown

Service	Config	Free Tier?	Cost
EC2	t3.micro (1 vCPU, 1GB RAM)	Yes	\$0

Option 2: Single EC2 + Docker Compose (Absolute Cheapest)

Architecture

```
Single t3.micro EC2 instance (free)
  ↓
  Docker Compose:
  ├── Node.js server (port 3000)
  ├── Python FastAPI (port 8000)
  ├── PostgreSQL (SQLite alternative for demo)
  ├── Redis (for sessions/cache)
  ├── Kafka (lightweight, embedded)
  └── All on 1GB RAM + 20GB EBS
```

Monthly Cost (First 12 Months)

Service	Config	Cost
EC2	t3.micro (free tier)	\$0
EBS	20GB gp3 (1 x 20GB)	\$0

Option 3: Hybrid (Best Balance)

Architecture

2 × t3.small EC2 (1 for Node, 1 for Python) = ~\$25/mo

↓

RDS db.t3.micro (free year 1, ~\$50/mo after) = \$0 / \$50

↓

S3 bucket (5GB free tier) = \$0 / \$10

↓

LocalStack for DynamoDB/SQS testing (runs in EC2) = \$0

↓

Lambda for async OCR (free tier) = \$0

Period	EC2	RDS	S3	Other	Total
Months 1–12	\$25	\$0	\$0	\$5	\$30–40
Month 13+	\$25	\$50	\$10	\$5	\$90–100

Quick Wins: Cut Costs Further

1. Skip NAT Gateway → Use Elastic IP (~\$32/mo saved)

```
# In Terraform: don't create a NAT; use Elastic IP on EC2
resource "aws_eip" "node" {
  instance = aws_instance.node.id
  domain   = "vpc"
  tags     = { Name = "nbe-demo-eip" }
}
# Cost: $0 while attached to running instance
```

2. Skip Managed Kafka → Use Lightweight Job Queue

```
# Instead of MSK, use SQS FIFO (free tier-eligible)
import boto3
sqs = boto3.client('sqs')

# For async OCR, Lambda + SQS
sqs.send_message(
```

Recommended Demo Setup (Best for Shipping Features Fast)

Option: Single EC2 t3.micro + Docker Compose + S3 + RDS micro

```
EC2 t3.micro (free tier, 1 year)
├── Node.js (port 3000)
├── Python FastAPI (port 8000)
├── PostgreSQL 15 (1GB, via container)
├── Redis 7 (in-memory, via container)
└── Kafka (lightweight, for events)

+ Elastic IP (free while attached)

├── S3 (free tier: 5GB + 20K PUTs)
├── RDS Postgres db.t3.micro (free year 1) | ← optional: if you want managed DB
└── Lambda (free tier: 1M invokes/mo)      | ← for serverless OCR
```

Exact Monthly Costs

Year 1 (Free Tier)

Launch Steps (30 min total)

1. Create AWS Free Tier Account

```
# Go to https://aws.amazon.com/free/  
# Sign up → Verify email → Add payment method  
# Set up billing alerts (CloudWatch)
```

2. Launch EC2 Instance

```
# Via AWS Console or CLI  
aws ec2 run-instances \  
  --image-ids ami-0c55b159cbfafa1f0 \  # Amazon Linux 2  
  --instance-type t3.micro \  
  --key-name my-key \  
  --security-groups default \  
  --tag-specifications 'ResourceType=instance,Tags=[{Key=Name,Value=nbe-demo}] '  
  
# Allocate Elastic IP (free!)  
aws ec2 allocate-address --domain vpc --output text | awk '{print $1}' > eip-id.txt  
aws ec2 associate-address --instance-id <INSTANCE_ID> --allocation-id $(cat eip-id.txt)
```

What Works at This Cost

Feature	Status	Notes
Multi-tenant document repo	✓	S3 or local EBS
Web UI (Node Express)	✓	Runs in container
API Gateway (Python FastAPI)	✓	Full /api/v1/* surface
Full-text search	✓	Postgres FTS or DynamoDB scan
OCR (Tesseract)	✓	Local or Lambda Textract
Workflows (Temporal/Zeebe)	⚠	Simplified local scheduler
Real-time notifications	✓	SNS/SQS (free tier)
Mobile app (Expo)	✓	Connects to /api/v1
RBAC	✓	Built-in

Cost Comparison Matrix

Tier	Compute	Database	Storage	Events	Total Year 1	Total Year 2
Serverless (this guide)	EC2 free	RDS free	S3 free	Lambda free	\$60–120	\$270–350
Managed K8s (Option 1)	EKS \$150	RDS \$200	S3 \$50	MSK \$400	\$1,800	\$1,800
Full SaaS (prod-ready)	ECS \$300	RDS \$500	S3 \$100	Lambda \$200	\$3,600	\$3,600

⚠ Limitations (Know Before You Ship)

Limitation	Workaround
t3.micro: 1 vCPU, 1GB RAM (tight)	Upgrade to t3.small (\$12/mo) for room
Single AZ (not HA)	Add second instance + load balancer (\$30/mo more)
Local Postgres in container (no managed backup)	Use RDS db.t3.micro (free year 1)
No auto-scaling	Manual: stop/start or scale instance size
100GB/mo free egress → overages \$0.09/GB	Use CloudFront (\$0.085/GB cheaper)



Demo Script (2 Hours)

```
# Terminal 1: Start stack
docker-compose -f docker-compose.prod-demo.yml up

# Terminal 2: Run demo
# 1. Login: nbe-demo.com/login (admin/admin123)
# 2. Upload doc: "Invoice_2026.pdf"
# 3. Watch OCR: Document→OCR Text (live Tesseract)
# 4. Classify: "Invoice" tag applied (AI)
# 5. Workflow: Maker→Checker→Approved (RBAC)
# 6. Search: "Invoice 2026" → FTS results
# 7. Export: ZIP download via S3
# 8. Mobile: Scan with Expo app, auto-upload
# 9. API: curl https://nbe-demo.com/api/v1/documents
# 10. Audit: View all actions in admin panel

# Show costs: aws ce get-cost-and-usage ...
```

Next: Production Migration

Once demo is approved, scale to:

- **Month 13:** Add second EC2 (HA) + RDS multi-AZ → ~\$100/mo
- **Month 15:** Migrate to EKS (1 cluster, 3 nodes) → ~\$300/mo
- **Month 18:** Add MSK Kafka → ~\$700/mo
- **Month 24:** Full multi-region SaaS → ~\$2,000/mo

Each step is backward-compatible; no code changes needed.